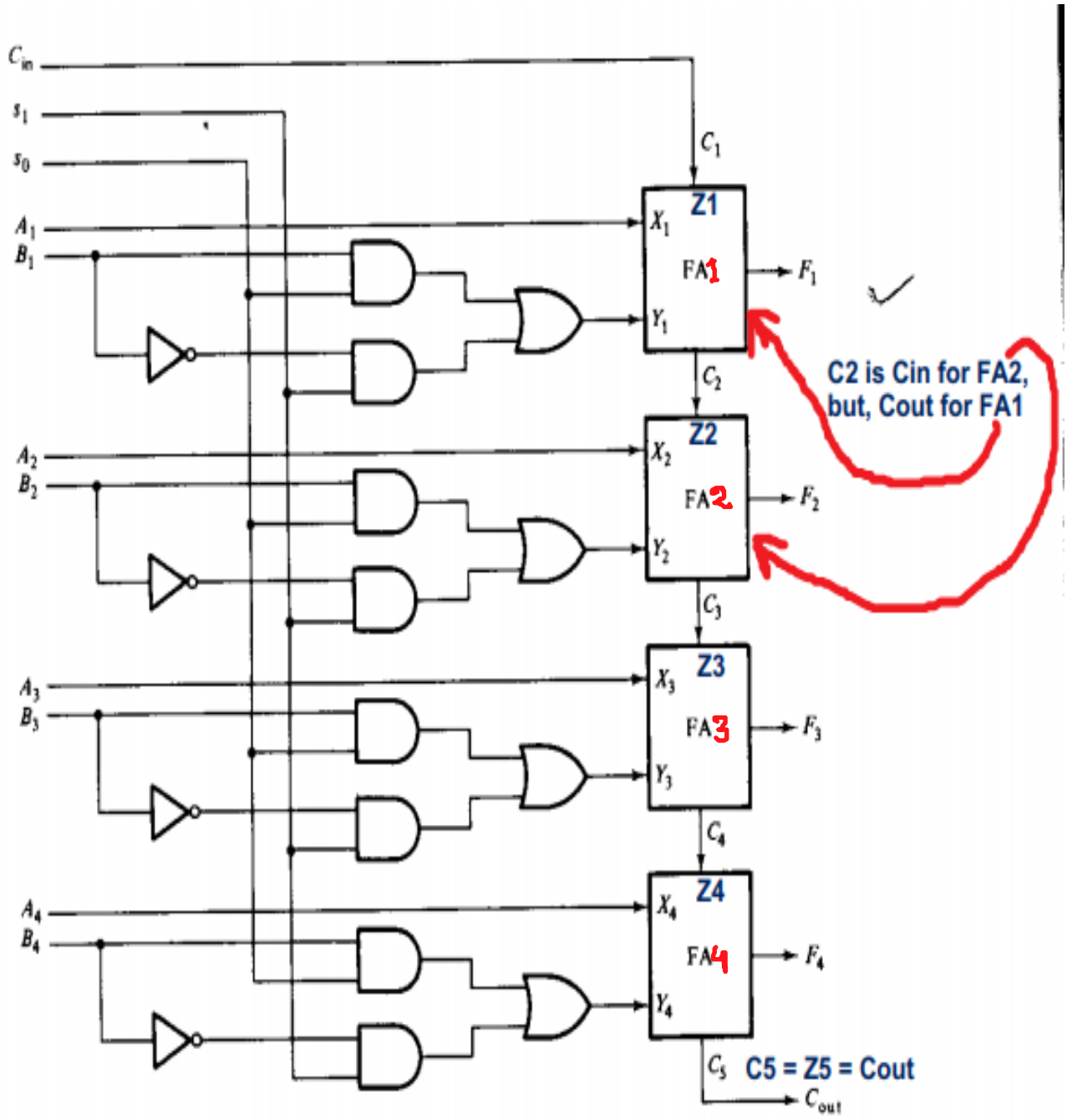
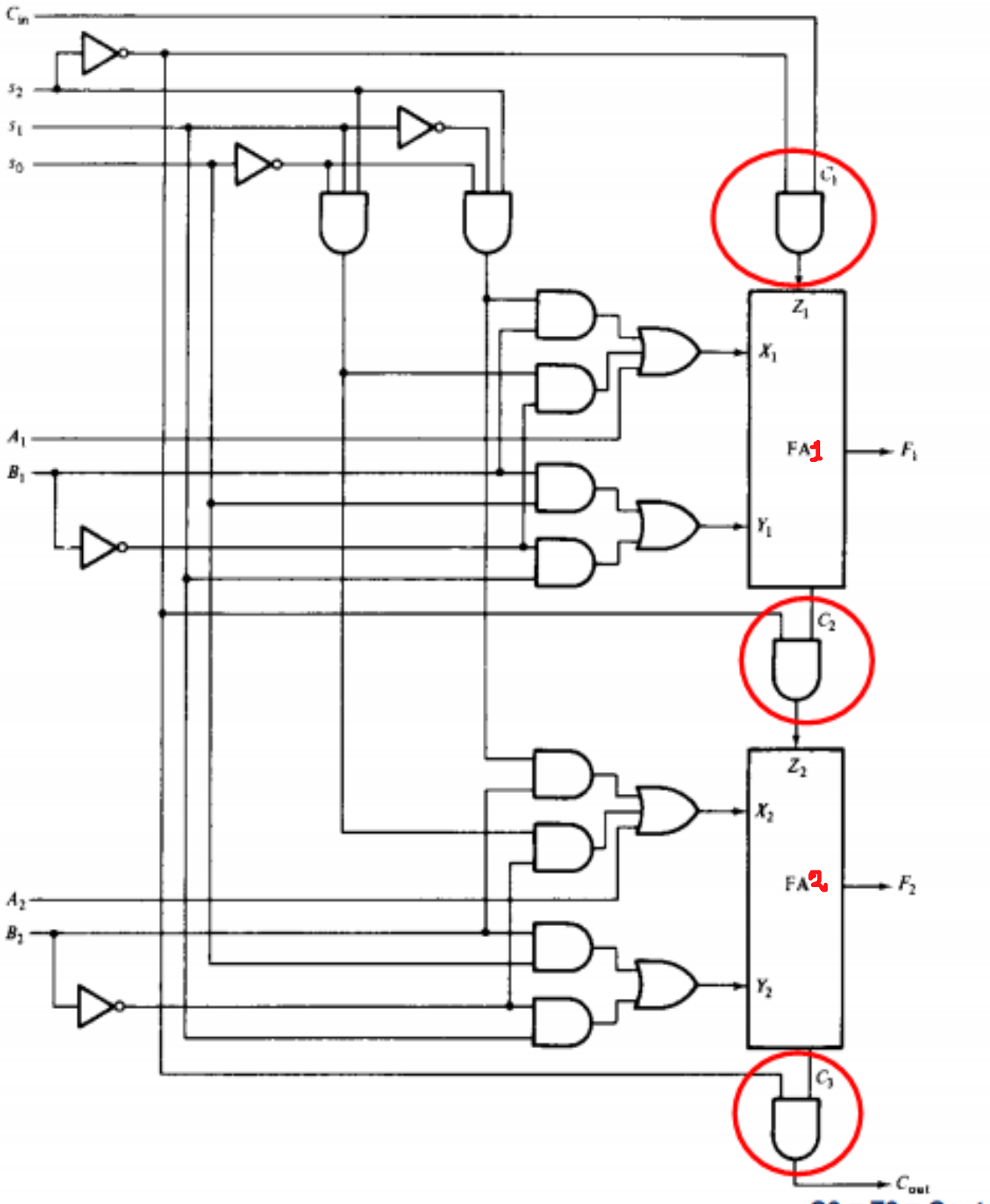

C_{in}
S_1
S_0
A_1
B_1
A_2
B_2
A_3
B_3
A_4
B_4
C_1
Z1
X_1
FA1
Y_1
F_1
C_2
Z2
X_2
FA2
Y_2
F_2
C_3
Z3
X_3
FA3
Y_3
F_3
C_4
Z4
X_4
FA4
Y_4
F_4
C_5
C_{out}
C2 is Cin for FA2, but, Cout for FA1
C5 = Z5 = Cout



C3 = Z3 = Cout

Figure 9-13 Logic diagram of arithmetic logic unit (ALU)

TABLE 9-3 Logic operations in one stage of arithmetic circuit

s_2	s_1	s_0	X_i	Y_i	C_i	$F_i = X_i \oplus Y_i$	Operation	Required operation
1	0	0	A_i	0	0	$F_i = A_i$	Transfer A	OR
1	0	1	A_i	B_i	0	$F_i = A_i \oplus B_i$	XOR	XOR
1	1	0	A_i	B_i'	0	$F_i = A_i \odot B_i$	Equivalence	AND
1	1	1	A_i	1	0	$F_i = A_i'$	NOT	NOT

XNOR

The value of X_i is always equal to input A_i. Table 9-3 shows the four logic operations obtained when a third selection variable $s_2 = 1$. This selection variable forces C_i to be equal to 0 while s_1 and s_0 choose a particular value for Y_i. The four logic operations obtained by this configuration are transfer, exclusive-OR, equivalence, and complement. The third entry is the equivalence operation because:

$$A_i \oplus B_i' = A_i B_i + A_i' B_i' = A_i \odot B_i$$

The last entry in the table is the NOT or complement operation because:

$$A_i \oplus 1 = A_i'$$